

Extended Essay Official Proposal

Due Date: February W4, 2026

Purpose: To finalize your research question, outline your plan, and show readiness to move into drafting.

Student Name: Mykhailo Prysiashniuk

Subject Area (choose one): Machine Learning(ML)

1. Working Title

Provide a clear and tentative title for your essay.

An experimental research and analysis on how effectively do different jailbreaking techniques, the 'roleplay person', the 'homoglyph encoding' and the "image input", break the policy of LLM, such as GLM-4.7.

2. Research Question (Finalized)

Write your research question in full, phrased as a question.

To which extent can the three distinct prompt injection techniques, the 'roleplay person', the 'homoglyph encoding', and the 'image input', overcome the safety policy of the GLM-4.7 LLM, through a Python framework, and what do the results imply for the design of effective AI defense mechanisms?

3. Rationale

(200–250 words)

- Why did you choose this subject and topic?
- Why is this question significant academically?
- How does it connect to your personal interests or future studies?

Computer science is the subject I am fond of - I want to study it in the university and connect it with my future, as I am thinking of professions of a software or AI developer, or others connected to coding. The problem of AI jailbreaking is inspirational for me, as it combines coding, which is my strength and area of interest, AI, which is the main center of attention in the modern world and takes more and more work from human, and security, which requires decent upgrades with release of new technologies.

This question is important for the world because it allows to hold an experiment, analyse and compare different techniques, and suggest solutions or improvements to the LLMs. Meanwhile the question about the security of AI is significant as online safety is a developing area, in which AI has several policy leakages that can lead to unpleasant and harmful

consequences for a person or society, as it can be used for creating biases or getting secret/confidential information.

It will give me a knowledge about Learning Language Models, how to work with them and how to test them; an opportunity to apply my programming abilities to a real project; shape my understanding of the modern technologies.

4. Initial Research & Sources

- List at least 5 key sources (books, articles, primary data, etc.).
- Indicate how each source supports your research.

Source	Type (Primary/ Secondary)	How it helps answer your RQ
https://huggingface.co/zai-org/GLM-4.7	Primary	Documentation on how to use the LLM GLM-4.7
https://www.promptfoo.dev/docs/red-team/	Primary	The main documentation on homoglyph, image techniques with overall information about different attacks
https://meetcyber.net/when-role-play-goes-rogue-prompt-injection-bypasses-in-llms-ca22913e06f0	Primary	Roleplay person method information
https://arxiv.org/pdf/2507.21820	Secondary	Additional information about jailbreaking techniques
OpenAI about prompt injection: https://openai.com/index/prompt-injections/	Secondary	Understanding of prompt injection
Prompt injection detection - https://openai.github.io/openai-guardrails-python/ref/checks/prompt_injection_detection/#results https://www.salesforce.com/blog/prompt-injection-detection/	Secondary	How to avoid prompt injections

5. Methodology / Approach

(150–200 words)

- How will you investigate your question? (e.g., textual analysis, data analysis, experiment, case study, etc.)
- What is your intended structure?
- How will you ensure academic integrity (proper citation, originality)?

I am intending to hold a digital experiment. The core methodology involves a Python-based framework that will create the Base 64 obfuscation and Roleplay person attacks, apply them to the LLM, record and visualise outcomes of the attacks.

My intended structure of the essay is: introduction + information and definitions; explanations about the key terms by using the sources; predictions about the results based on the LLM documentation; the experiment - explanation of how it was held (a python script, the process, the results visualization). After that I would explain what do these results suggest and how could we make a better protection system for the LLM.

Finally, I would add all the necessary details, such as conclusion, references. I would ensure that I add references to all information from external sources. I will also write all the code from scratch and apply the techniques by myself, using only downloaded LLM.

6. Anticipated Challenges

- What difficulties do you expect (access to sources, data limitations, scope)?
- How do you plan to address them?

One of the problems is that I cannot put any unethical information into the Extended Essay; meanwhile jailbreaking requires a prompt that breaks the policy of the LLM. So I have to use prompts that I cannot show or find a specific prompt that is against Mistral policy but acceptable, or find any other ways.

There might be difficulties with downloading the software. However, they should be manageable.

One of the problems is that AI is still developing and there might not be much information about such topic as injections; however, I want to base the essay on my experiment.

There might be a problem where some software/tool is not free.

7. Proposed Timeline (November 2025 – September 2026)

- November – February: PHASE 1

- March – April: **PHASE 2**
- May – September: **PHASE 3**

Aligned Deadlines (Nov. 2025 – Sep 2026)

Section	Task	Deadline - Non-Negotiable
Proposal	Research question, rationale, sources, methodology	February/W4
Introduction	Draft context, RQ, scope, methodology	May/W1
Body (Full Draft)	All sections written	May/W2-3
Conclusion	Draft and integrate	May/W4
Draft Submission	Full essay draft submitted for feedback	June 15 th , 2026
Supervisor Feedback	June 2026 (end of DP1)	
Revisions & Editing	Summer break	June–August 2026
Final Submission	EE + RPF (500 words)	Third week of September 2026 (DP2)

Student's signature:

u. Thomas

Supervisor's signature:

Ty